

Folded SBP Operators for Spherical Symmetry: Exact Origin Regularization, SAT Wave Dynamics, and Stability Diagnostics

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This report documents the implemented methodology in `SphericalSBPOperators.jl` for constructing radial summation-by-parts (SBP) operators on $[0, R]$ from mirrored Cartesian operators on $[-R, R]$. The method enforces parity by construction, uses a metric-weighted diagonal mass, and defines discrete divergence through the SBP identity. A key implementation feature is an exact near-origin repair of the folded even-gradient operator in rational arithmetic, coupled to divergence constraints, followed by exact/near-exact verification. The resulting operators are used in a first-order radial wave system with characteristic SAT boundary conditions (absorbing, reflecting, Dirichlet). We present the full step-by-step construction, imposed constraints, and rationale, then derive semidiscrete energy-rate identities and stability conditions for each boundary closure. Representative numerical results from the current code path show machine-level SBP residuals, closure-localized polynomial errors, spectral stability (no exponentially growing modes), and near-zero energy drift for reflecting SAT with implicit midpoint time integration.

I. PROBLEM STATEMENT AND DESIGN REQUIREMENTS

We target radial operators on $r \in [0, R]$ for

$$\text{Div}_p(u) = \frac{1}{r^p} \frac{d}{dr} (r^p u), \quad p \in \{0, 1, 2\}, \quad (1)$$

with $p = 2$ (spherical), $p = 1$ (cylindrical), and $p = 0$ (Cartesian radial reduction).

The implementation requirements are:

1. one half-grid degree-of-freedom layout for both even/scalar and odd/flux fields;
2. parity consistency at the origin enforced by extension maps, not ad hoc row edits;
3. metric-consistent discrete SBP structure

$$H_{\text{sph}} D + G_{\text{even}}^T H_{\text{sph}} = B; \quad (2)$$

4. regular origin treatment consistent with $\lim_{r \rightarrow 0} \text{Div}_p(u) = (p+1)u'(0)$ for odd regular u ;
5. practical stability support for wave propagation with characteristic SAT boundaries.

II. CONTINUUM MODEL AND PARITY CONDITIONS

A. Parity

At $r = 0$, parity is a symmetry requirement:

$$\phi(-r) = \phi(r) \quad (\text{even scalar-like field}), \quad (3)$$

$$u(-r) = -u(r) \quad (\text{odd radial flux-like field}), \quad (4)$$

hence $u(0) = 0$.

B. Removable singularity at the origin

For odd regular $u(r) = a_1 r + a_3 r^3 + \dots$,

$$\lim_{r \rightarrow 0} \text{Div}_p(u) = (p+1)a_1 = (p+1)u'(0). \quad (5)$$

This condition is explicitly imposed in the discrete divergence first row.

C. First-order radial wave system

The implemented semidiscrete wave state is (Π, Ξ) with

$$\partial_t \Pi = D \Xi, \quad \partial_t \Xi = G_{\text{even}} \Pi. \quad (6)$$

At the outer boundary $r = R$, characteristic variables are

$$w_{\text{in}} = \Pi + \Xi, \quad w_{\text{out}} = \Pi - \Xi. \quad (7)$$

SAT penalties are applied on the RHS (not via state projection) according to the chosen boundary condition.

III. FOLDED SBP CONSTRUCTION ON $[0, R]$

A. Step 1: mirrored Cartesian operator

Let the full grid on $[-R, R]$ be

$$\mathbf{x}^{\text{full}} = (x_1, \dots, x_M), \quad x_1 = -R, \quad x_M = R, \quad M = 2N+1. \quad (8)$$

From `SummationByPartsOperators.jl`, build a Cartesian first-derivative operator G^{full} and full mass matrix H^{full} . Matrix extraction supports:

1. direct dense conversion when square matrix conversion exists,
2. probing columns by applying the operator to basis vectors.

B. Step 2: folding maps and half-grid

Define the nonnegative half-grid indices

$$\mathcal{I}_+ = \{j \mid x_j \geq 0\}, \quad \mathbf{r} = (r_1, \dots, r_{N_h}), \quad r_1 = 0, \quad r_{N_h} = R. \quad (9)$$

Build sparse maps

$$\mathcal{R} \in \mathbb{R}^{N_h \times M} \quad (\text{restriction}), \quad (10)$$

$$E_{\text{even}} \in \mathbb{R}^{M \times N_h} \quad (\text{even extension}), \quad (11)$$

$$E_{\text{odd}} \in \mathbb{R}^{M \times N_h} \quad (\text{odd extension}). \quad (12)$$

For each full-grid point x_j , let $\sigma(j)$ satisfy $r_{\sigma(j)} = |x_j|$:

$$(E_{\text{even}})_{j,\sigma(j)} = 1, \quad (13)$$

$$(E_{\text{odd}})_{j,\sigma(j)} = \text{sign}(x_j), \quad \text{sign}(0) = 0. \quad (14)$$

The odd extension therefore enforces zero at the origin by construction.

C. Step 3: folded derivatives

$$G_{\text{even}} = \mathcal{R} G^{\text{full}} E_{\text{even}}, \quad (15)$$

$$G_{\text{odd}} = \mathcal{R} G^{\text{full}} E_{\text{odd}}. \quad (16)$$

G_{even} maps even fields to odd derivatives; G_{odd} maps odd fields to even derivatives.

D. Step 4: metric mass and boundary operator

Even-function folding gives

$$H_{\text{cart,half}} = \frac{1}{2} E_{\text{even}}^T H^{\text{full}} E_{\text{even}}. \quad (17)$$

Metric weighting produces

$$H_{\text{sph}} = H_{\text{cart,half}} \text{diag}(r_1^p, \dots, r_{N_h}^p). \quad (18)$$

The boundary matrix has only the outer contribution:

$$B = \text{diag}(0, \dots, 0, R^p). \quad (19)$$

E. Step 5: exact coupled near-origin repair of G_{even}

After direct folding, some near-origin rows may retain nonzero first-column coupling $(G_{\text{even}})_{i1} \neq 0$ ($i \geq 2$), which contaminates odd/even decoupling. The code repairs these rows in exact rational arithmetic (`Rational{BigInt}` path).

Define

$$\mathcal{I}_c = \{i \in \{2, \dots, N_h\} \mid (G_{\text{even}})_{i1} \neq 0\}. \quad (20)$$

For rows in $\mathcal{I}_c$, unknown coefficients are solved on a stencil $\mathcal{S} \subset \{2, \dots, N_h\}$ under two coupled constraint families:

(a) *Even-gradient exactness constraints* For $m = 0, \dots, q_b$ (with inferred Cartesian boundary exactness q_b),

$$\sum_{j \in \mathcal{S}} g_{ij} r_j^{2m} = \frac{d}{dr} (r^{2m}) \Big|_{r=r_i} \quad (\text{fixed contribution from non-stencil columns}) \quad (21)$$

(b) *Divergence consistency constraints* For odd degrees $d \in \{1, 3, \dots, q - p\}$ admitted by inferred interior order q ,

$$\frac{B_{ii} r_i^d - \sum_j (G_{\text{even}})_{ji} H_{jj} r_j^d}{H_{ii}} = (p + d) r_i^{d-1}, \quad i \in \mathcal{I}_c. \quad (22)$$

The linear system is solved exactly (Gauss-Jordan over rationals), then verified exactly. If no exact solution exists for the requested constraint set, the implementation increases stencil width and, if still necessary, relaxes highest odd-degree divergence constraints until consistency is reached. This is a deliberate robustness strategy, not an accidental fallback.

F. Step 6: divergence construction and origin closure

For rows $i \geq 2$:

$$H_{\text{sph}} D = B - G_{\text{even}}^T H_{\text{sph}}, \quad D_{ij} = \frac{(B - G_{\text{even}}^T H_{\text{sph}})_{ij}}{(H_{\text{sph}})_{ii}}. \quad (23)$$

At $r = 0$, for $p > 0$, $(H_{\text{sph}})_{11} = 0$ and SBP does not constrain row 1. The code imposes

$$D_{1,:} = (p + 1)(G_{\text{odd}})_{1,:}, \quad (24)$$

which enforces $(Du)(0) = (p + 1)u'(0)$ for odd u .

G. Step 7: canonical-to-physical scaling

Construction is performed on a canonical uniform grid with $\Delta r = 1$ and then scaled to requested physical R . With $r = \rho \hat{r}$:

$$G_{\text{even}}, G_{\text{odd}}, D \mapsto \rho^{-1} (G_{\text{even}}, G_{\text{odd}}, D), \quad (25)$$

$$H_{\text{sph}} \mapsto \rho^{p+1} H_{\text{sph}}, \quad (26)$$

$$B \mapsto \rho^p B. \quad (27)$$

IV. WHAT CONDITIONS ARE IMPOSED AND WHY

1. **Origin parity:** odd fields satisfy $\Xi(0) = 0$, enforced in state and RHS space, to keep evolution inside the odd subspace and avoid origin-mode leakage.
2. **Origin divergence regularity:** $D_{1,:} = (p + 1)(G_{\text{odd}})_{1,:}$ enforces the removable singularity limit.

3. **SBP structure:** D is derived from $H_{\text{sph}}D + G_{\text{even}}^T H_{\text{sph}} = B$ for $i \geq 2$, ensuring summation-by-parts consistency with the metric mass.
4. **Near-origin decoupling repair:** exact coupled constraints remove folded first-column contamination while preserving polynomial exactness targets.
5. **Characteristic SAT closure:** boundary conditions are imposed weakly on RHS to control energy (dissipative for absorbing, conservative cancellation for reflecting/Dirichlet).

V. VERIFICATION AND DIAGNOSTICS WORKFLOW

A. Closure-aware safe interior

Boundary closure widths are estimated from row-pattern mismatch (and optional coefficient mismatch) relative to an interior reference row. Right closure is made conservative by combining pattern-based width with SBP metadata (`coefficients.right_weights`) when available.

Safe interior indices are then

$$\mathcal{I}_{\text{safe}} = \{1 + w_{\text{left}}, \dots, N_h - w_{\text{right}}\}, \quad (28)$$

with additional conservative offset on the left for divergence checks.

B. Validation quantities

For monomial tests up to selected degree:

1. weighted quadrature exactness using H_{sph} ;
2. gradient errors for even monomials under G_{even} ;
3. divergence errors for odd monomials under D ;
4. SBP residual norms of $R_{\text{SBP}} = H_{\text{sph}}D + G_{\text{even}}^T H_{\text{sph}} - B$ (full and excluding origin row).

Reported metrics include global max, safe-interior max, and near-boundary max.

C. Extended diagnostics

The `diagnose(...)` path additionally checks:

1. grid symmetry/uniformity and folded-grid consistency;
2. algebraic consistency of $(\mathcal{R}, E_{\text{even}}, E_{\text{odd}})$ maps;
3. direct-vs-reconstructed folded operator differences;

4. localization of polynomial-error hotspots (origin/mid/boundary);
5. interpreted conclusions (folding issue, closure dominance, origin-dominant divergence, etc.).

VI. WAVE-SYSTEM DISCRETIZATION AND BOUNDARY CONDITIONS

A. Semidiscrete system

The implemented RHS is

$$\dot{\Pi} = D\Pi + \text{SAT}_{\Pi}, \quad (29)$$

$$\dot{\Xi} = G_{\text{even}}\Pi + \text{SAT}_{\Xi}, \quad (30)$$

with optional origin projection $\dot{\Xi}(0) = 0$.

B. Characteristic SAT penalties at $r = R$

Define residual ρ and penalties:

$$\dot{\Pi}_N \leftarrow \dot{\Pi}_N - \sigma_{\Pi} H_{NN}^{-1} \rho, \quad (31)$$

$$\dot{\Xi}_N \leftarrow \dot{\Xi}_N - \sigma_{\Xi} H_{NN}^{-1} \rho. \quad (32)$$

Implemented cases:

$$\text{absorbing: } \rho = w_{\text{in}}, \quad \sigma_{\Pi} = \sigma_{\Xi} = B_{NN}/2, \quad (33)$$

$$\text{reflecting: } \rho = w_{\text{in}} - w_{\text{out}} = 2\Xi_N, \quad \sigma_{\Pi} = B_{NN}/2, \quad \sigma_{\Xi} = 0, \quad (34)$$

$$\text{Dirichlet: } \rho = w_{\text{in}} + w_{\text{out}} = 2\Pi_N, \quad \sigma_{\Pi} = 0, \quad \sigma_{\Xi} = B_{NN}/2. \quad (35)$$

Aliases `:radiative` and `:reflective` map to absorbing/reflecting.

C. Jacobian and integration

The package provides both matrix-state and vector-state ODE forms, analytic Jacobian, and sparse Jacobian prototype for implicit midpoint solves.

VII. STABILITY ANALYSIS

A. Semidiscrete energy identity

Define

$$E = \frac{1}{2} (\Pi^T H_{\text{sph}} \Pi + \Xi^T H_{\text{sph}} \Xi). \quad (36)$$

For the interior part (without SAT),

$$\dot{E}_{\text{int}} = \Pi^T H_{\text{sph}} D \Xi + \Xi^T H_{\text{sph}} G_{\text{even}} \Pi = \Pi^T (B + R_{\text{SBP}}) \Xi, \quad (37)$$

where $R_{\text{SBP}} = H_{\text{sph}}D + G_{\text{even}}^T H_{\text{sph}} - B$. Hence

$$\dot{E}_{\text{int}} = \underbrace{B_{NN}\Pi_N\Xi_N}_{\text{boundary flux}} \underbrace{\Pi^T R_{\text{SBP}}\Xi}_{\text{SBP residual term}}. \quad (38)$$

B. Boundary-condition-specific energy behavior

Reflecting SAT SAT contributes $-B_{NN}\Pi_N\Xi_N$, canceling boundary flux:

$$\dot{E}_{\text{refl}} = \Pi^T R_{\text{SBP}}\Xi. \quad (39)$$

Thus reflecting energy is conservative up to SBP residual and origin-degeneracy effects.

Dirichlet SAT An analogous cancellation occurs, again giving

$$\dot{E}_{\text{Dir}} = \Pi^T R_{\text{SBP}}\Xi. \quad (40)$$

Absorbing SAT Using $\rho = w_{\text{in}} = \Pi_N + \Xi_N$ with symmetric penalties:

$$\dot{E}_{\text{abs}} = -\frac{B_{NN}}{4} (w_{\text{in}}^2 + w_{\text{out}}^2) \Pi^T R_{\text{SBP}}\Xi, \quad (41)$$

which is dissipative apart from the SBP residual term.

C. Spectral checks

The stability suite (`scripts/StabilityAnalysis.jl`) computes eigenvalues of the closed block operator

$$L = \begin{bmatrix} 0 & D \\ G_{\text{even}} & 0 \end{bmatrix} \text{SAT terms}, \quad (42)$$

flags modes with $\text{Re}(\lambda) > \text{tolerance}$, estimates RK4 timestep limits from the full spectrum, and inspects near-null modes for checkerboard signatures.

D. Runtime consistency checks

The implementation also probes the runtime RHS operator by basis application and compares it to the analytic Jacobian operator. For reflecting runs it evaluates:

1. direct RHS energy rate $\dot{E}$,
2. flux and SBP residual decomposition,
3. skew defect $S = A^T H_{\text{blk}} + H_{\text{blk}} A$ (full/no-origin/safe blocks),
4. quadratic-form consistency $\dot{E} \approx \frac{1}{2} U^T S U$.

VIII. REPRESENTATIVE RESULTS FROM CURRENT IMPLEMENTATION

A. Operator validation (accuracy_order=6, N=64, R=1, p=2)

These data are closure-dominated globally while safe-interior errors are near machine precision, consistent with high-order SBP boundary closure behavior.

B. Spectral and time-integration stability diagnostics

C. Reflecting SAT drift diagnosis

For a representative reflecting test ($R=8$, $\text{dr}=0.2$, `ImplicitMidpoint`, short horizon):

- runtime-vs-analytic operator mismatch: exactly zero in reported max norm;
- no-origin skew defect: 1.42×10^{-14} ;
- $\max |\dot{E}|$ over early diagnostic states: 6.94×10^{-17} (Gaussian) and 6.07×10^{-18} (compact bump).

This supports the code-level conclusion that semidiscrete reflecting drift is negligible in these settings, and observed long-time drift (when present) is typically time-integration or data-consistency related.

IX. ALGORITHM SUMMARY (IMPLEMENTATION ORDER)

1. Build full-grid Cartesian SBP operators on $[-R, R]$.
2. Build $\mathcal{R}, E_{\text{even}}, E_{\text{odd}}$ and half grid $\mathbf{r}$ with explicit origin check.
3. Fold to obtain $G_{\text{even}}, G_{\text{odd}}$.
4. Form $H_{\text{cart, half}}$, then metric mass $H_{\text{sph}} = H_{\text{cart, half}} \text{diag}(r^p)$ and outer-boundary B .
5. Identify contaminated near-origin rows of G_{even} and repair them by exact coupled constraints (Sec. III E).
6. Construct divergence from SBP for rows $i \geq 2$; set row 1 to $(p+1)(G_{\text{odd}})_{1,:}$.
7. Optionally scale canonical operators to requested physical radius.
8. Validate (quadrature, polynomial derivative/divergence, SBP residuals) with closure-aware safe interior.
9. Evolve wave system with SAT boundary condition and parity-preserving origin RHS constraint.

TABLE I. Representative validation diagnostics from current code execution.

Quantity	Value
Closure widths ($w_{\text{left}}, w_{\text{right}}$)	(7, 6)
Safe interior count	52
$\ R_{\text{SBP}}\ _{\infty}$ (full)	1.11×10^{-16}
$\ R_{\text{SBP}}\ _{\infty}$ (excluding origin row)	1.11×10^{-16}
Gradient error, degree 4: global / safe / near-boundary	6.19×10^{-5} / 1.78×10^{-15} / 6.19×10^{-5}
Gradient error, degree 6: global / safe / near-boundary	8.53×10^{-4} / 1.78×10^{-15} / 8.53×10^{-4}
Divergence error, degree 3: global / safe / near-boundary	3.16×10^{-4} / 7.11×10^{-15} / 3.16×10^{-4}

10. Run spectral and runtime energy diagnostics for stability assurance.

X. REPRODUCIBILITY NOTES

Representative runs in this report were produced from the current package code using the following entry points:

1. `validate(ops)` and `diagnose(ops)` in `src/validation.jl` and `src/diagnostics.jl`,
2. `solve_wave_ode` and `diagnose_reflecting_sat_energy_drift` in `src/wave_solver.jl`,
3. `spectral_analysis`, `test_pole_stability`, and `run_stability_suite` in `scripts/StabilityAnalysis.jl`.

XI. CONCLUSIONS

The implemented method provides a coherent, publication-grade SBP framework for radial problems:

1. parity-compatible folding from mirrored Cartesian grids,
2. exact near-origin repair for folded even-gradient consistency,

3. SBP-consistent divergence with explicit removable-singularity origin closure,

4. characteristic SAT boundary enforcement with clear energy behavior,

TABLE II. Representative stability metrics.

Metric	Value
Reflecting-spectrum max $\text{Re}(\lambda)$	2.47×10^{-14}
Spectral radius $\rho(L)$	2.52×10^3
RK4 Δt_{max} from full spectrum	1.123×10^{-3}
Reflecting run, RK4 ($\Delta t = 5 \times 10^{-4}$): E_f/E_0	0.999999999958
Reflecting run, ImplicitMidpoint ($\Delta t = 5 \times 10^{-4}$): E_f/E_0	0.999999999999

5. integrated spectral and runtime diagnostics that localize instability sources.

The primary practical lesson is that closure-aware interpretation is essential: global polynomial errors can be boundary-local while interior consistency remains at machine precision.

ACKNOWLEDGMENTS

This report reflects the implementation in `SphericalSBPOperators.jl` at the current workspace state and is intended as a method paper draft for further refinement.

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